TOKAI Susano

RoboCup Maze 2026 Engineering Journal

Team Members

Name	Age	School	Address	Role	Main Achievements
Riku Harada	(15)	Tokai High School, Grade 2	Kakamigahara, Gifu	Overall Leader	FLL 2025 Japan Championship Winner
Yoshiharu Tsujimoto	(15)	Tokai High School, Grade 1	Nagoya, Aichi	Hardware	FLL 2025 Japan Championship Winner
Seigo Sobue	(13)	Tokai Junior High School, Grade 3	Nagoya, Aichi	Software	WRO 2024 Middle Category Japan Runner-up

Rule Analysis and Required Robot Performance Date: 2025.07.09

· Rescue as many victims as possible.

· Reliably read victim colors and recognize letters.

· Reliably identify floor colors, especially black and blue.

· Return to the start position and exit the maze.

· Clearly know where the robot has traveled.

· First, build a robot that can follow walls and complete one loop.

· Build the robot as small as possible (target: within 16 cm).

· An AI camera is essential for letters, floating tiles, floor colors, and self-localization.

· The competition time is not very long, so a certain level of speed is also required.

· Do not aim for a perfect score; use a strategy that earns points more reliably.

· The course uses only right angles, with no diagonal paths, and each tile is 30 cm x 30 cm.

· Obstacles exist. First, complete the missions on a flat course.

Milestones (Development Roadmap) Date: 2025.07.10

July

Analyze the rules and set the plan, strategy, and goals.

Control the robot using only motors and Arduino (*Prototype Robot V1).

Build a simple course (3 x 4 tiles).

August

Attach touch sensors and perform wall-following tests.

Attach ToF sensors and perform wall-following tests.

Attach a gyro sensor and perform wall-following tests.

Attach an AI camera and perform wall-following tests.

September

Detect victims with the AI camera (first step toward scoring points).

Identify victims using the AI camera.

October

Design a mechanism to drop rescue kits.

Run mock matches in competition format.

November

Adjust the robot and build Competition Robot No. 1.

December

Node tournament (regional tournament).

January 2026

Block tournament (Gifu District).

Build Robot No. 2.

February

Improve the completion level of Robot No. 2.

March

Participate in the Japan Championship.

Goals and Strategy

[Goal] Qualify for the World Championship (top 2 at Japan Open).

Strategy Overview

・Build multiple robots, including a spare robot.

・Select the robot that can score the highest among the available robots.

・Make full use of the AI camera and turn it into a strength of the robot.

・Consider introducing LiDAR.

・Improve the rescue accuracy for the “H” on floating tiles.

・Do not aim for a perfect score; focus on earning more points efficiently.

・Build a spare robot so development and testing can proceed in parallel.

・Use four-wheel drive.

・Make the robot smaller, especially from Robot No. 2 onward.

・Store the overall shape of the maze.

・Determine the optimal route while balancing remaining time and score.

Component Selection Study

AI Camera Candidates

・Husky Lens (JPY 7,000-10,000): Beginner to intermediate level; capable of letter recognition.

・M5Stack Unit V2 (JPY 7,000-9,000): Beginner to intermediate level.

・OpenMV Cam H7 PLUS (from JPY 13,000): Intermediate level; high performance but requires practice.

・Luxonis OAK-D / OAK-D Lite: Equipped with a depth camera; capable of advanced recognition.

Microcontroller Candidates

・Teensy 4.0 / 4.1 (JPY 6,000 / 7,000): High processing power; uses Teensyduino.

・STM32 F446RE: High performance.

· Raspberry Pi Pico: Low cost, but processing performance may be lower than Teensy.

Operation Check of Encoder Gear Motors Date: 2025.07.12

Motor selection: JGA25-371 (about JPY 2,000 each on Amazon)

Voltage: DC 6 V

Rotational speed: 100 rpm

Output: 0.14 W

Current: no-load 80-500 mA / max. 1.2 A (continuous) / 3.2 A (momentary)

Note: The current changes depending on the load.

Motor driver: TB6612FNG

Pin assignment for connection to the control microcontroller

Microcontroller Side (Teensy/Arduino)	Motor Side	Function / Notes
D5	PWMA	Motor A speed control (PWM)
D4	AIN2	Motor A direction control 2
D3	AIN1	Motor A direction control 1
D7	STBY	Standby pin (operates at High)
D9	BIN1	Motor B direction control 1
D10	BIN2	Motor B direction control 2
D11	PWMB	Motor B speed control (PWM)
GND	GND	Common ground
Vm (Battery)	VM	Motor power supply (up to 15 V)
Vcc (5V/3.3V)	VCC	Logic power supply

Test Item 1

Gradually increase the PWM value from 0 to 255 and observe the operating condition.

Results and Findings

· The motor generally starts moving at around PWM value 12.

· The minimum stable speed is about 15/255. There are individual differences; 15 is stable, but lower values are not stable.

· Using functions makes the program cleaner.

Example: rotateMotor(direction, speed, time, display) stopMotor(stop time)

・ Motor wiring and connection destinations

Cable Color	Connection	Function
Red	A01	Motor output
Black	GND	GND
Yellow	D2 (Arduino)	A phase (interrupt pin)
Green	D6 (Arduino)	B phase (not used this time)
Blue	5V (Arduino)	Encoder power supply
White	A02	Motor output

• Interrupt pins: Only D2 and D3 on Arduino.

(Interrupt pins are special pins that monitor signals so they can respond at any time.)

Test Item 2

Measure how many pulses are generated per one rotation.

Results and Findings

・ Encoder function: 13 pulses per motor rotation.

・ Gear ratio: 1:48

・ Pulses per tire rotation = 13 x 48 = 624 pulses.

・ Resolution: 360 degrees / 624 = 0.577 degrees per pulse.

・ Distance conversion: tire circumference = 65 mm x 3.14 = 204 mm.

・ Distance per pulse = 0.327 mm.

・ Pulses per 1 cm = 30.58 pulses.

Test Item 3

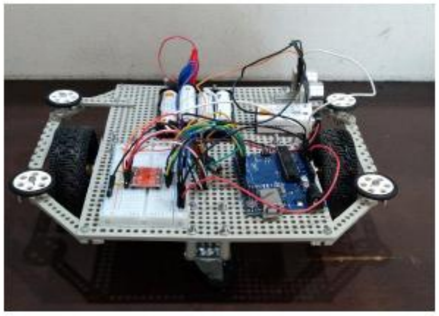
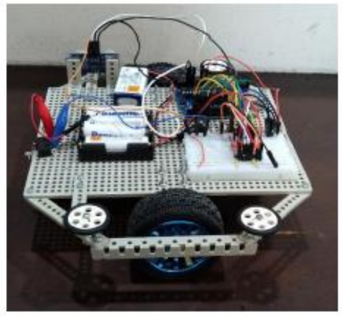
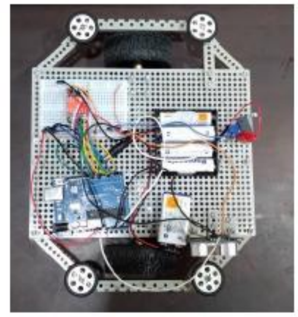
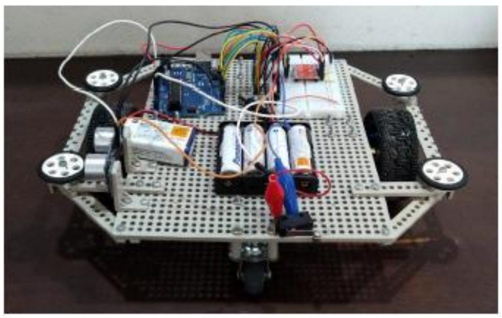
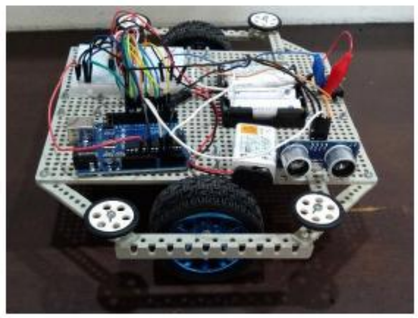
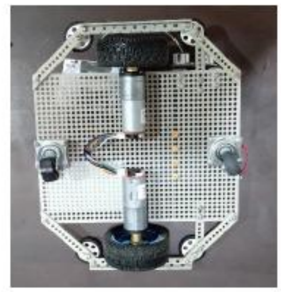
Rotate exactly one rotation and negative one rotation.

Results and Findings

・ It stops with a slight overshoot. This is a significant issue and the accuracy is poor.

・ Even when braking is applied, the overshoot remains because inertia is acting.

Prototype Robot V1 Development Date: 2025.07.21 to late August

Configuration

Microcontroller: Arduino Uno (compatible board)

Motor: JGA25-371 (100 rpm)

Motor driver: TB6612FNG

Battery: Eneloop AA x4 for motors, 9 V rectangular battery for Arduino

Sensors: Ultrasonic sensor (HC-SR04), touch sensor (simple microswitch)

Frame: Tamiya universal plate

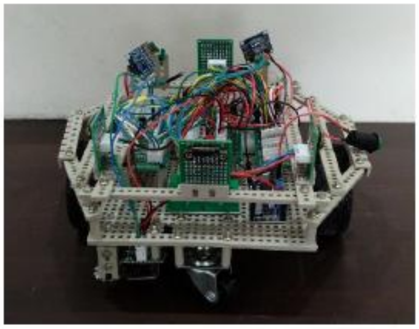
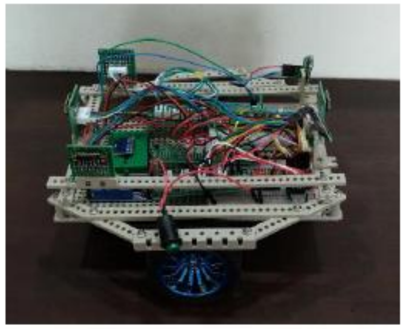
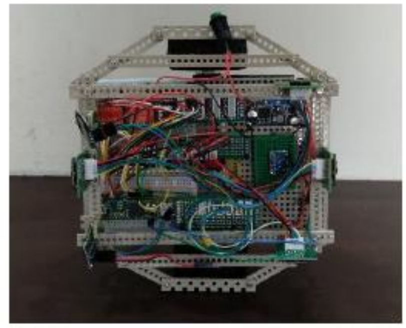
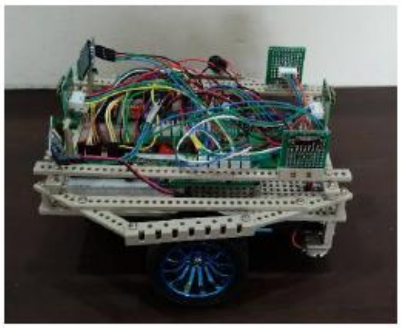
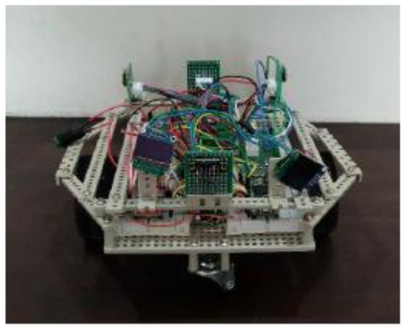
Completed Tasks

The robot completed one loop on a 4 x 3 course using the right-hand rule. It followed the right wall with an ultrasonic sensor and turned left when it hit the front wall.

Future Task

Reach the
floating tile

<u>Prototype Robot V2 Development Date: 2025.08.11 to late September</u>


<u>Configuration</u>

Microcontroller: Teensy 4.1

Motor: JGA25-371 (50 rpm)

Motor driver: TB6612FNG

Battery: Ni-MH batteries (Eneloop) AA x5 for motors, AA x4 for Teensy

Sensors: ToF sensors x4 (front, rear, left, and right), gyro sensor, color sensor

Lighting: LED

Display: OLED (SSD1306) x2

Frame: Tamiya universal plate

<u>Improvements from Prototype Robot V1 to V2</u>

· Microcontroller: Arduino to Teensy 4.1 (greatly improved processing performance).

· Motor: 6 V 100 rpm to 6 V 50 rpm (better low-speed control accuracy and higher torque).

· Slimmer body: width reduced from 26 cm to 21 cm.

· Wiring: Added XH connectors on a universal board to improve communication stability.

· Four ToF sensors enabled self-map generation and high-precision driving. Replacing the ultrasonic sensor with ToF sensors saved space and allowed control using only two Teensy pins via I2C communication.

· Interface: switches for motor/control/LED, two OLED displays, and a full-color LED.

· Added a color sensor.

Completed Tasks

· Reached the floating tile.
· Reached all areas of the map using the mapping program.
· Recorded the number of visits to every map tile.
· Returned to the start position.
· Read floor colors.

Issues and Countermeasures for Prototype Robot V2

· Braking accuracy was poor; the robot tilted and became unstable when stopping.
· Use of the gyro sensor was temporarily stopped because ToF values fluctuated when it was used. I2C communication congestion may be the cause, and countermeasures will be considered from Prototype V3 onward.
· Detect victims by installing an AI camera.
· Install a rescue-kit dropping mechanism.
· Create the frame and body using a 3D printer.
· Install touch sensors.
· Install LEDs for victim detection.
· Install a pause button.
· Supply stable 5 V power to the Teensy and AI camera.

Prototype Robot No. 3 Concept Date: 2025.08.17

・Aim to complete it between late September and mid-December.

・Aim for a completion level suitable for competition.

・Four-wheel drive.

・AI camera installed.

・LiDAR installed (SLAM).

・Raspberry Pi 5 installed for higher-level processing.

・Additional battery: 10,000 mAh.

・Rescue-kit dropping mechanism installed.

・Gyro: BNO08x or equivalent.

・Color sensors: use two AS7341 sensors.

・PCB design: create a dedicated board with KiCAD.

・Body: create with 3D printing.

・Tire improvement: improve grip and related performance.

Koriyama Exchange Meeting Technical Notes Date: 2025.08.30

Advice from TanoRobo (Mr. Miyazato)

■ Elements of a Strong Robot Learned at the Exchange Meeting

・High maintainability; it can be disassembled in about one minute.

・Use Fusion 360 proficiently.

・Gyro sensor: the BNO08x series is recommended for high accuracy.

・Color sensors: place one at the front and one at the rear; additional LEDs may not be necessary.

・Rescue-kit dropping mechanism: use one mechanism for both loading and pushing out.

・Rescue-kit material: brass, to prevent it from rolling too much.

・Battery: Li-Po 1500 mAh 20C, prioritizing weight reduction.

・Camera: keep it running continuously and use machine learning as well.

・LiDAR: use it to determine whether an obstacle is a wall.

■ Know-how for Custom-made Tires

・Tires should be large silicone tires with a shape that can climb over bumps, and the wheels should also be custom-made.

・Manufacturing method: create molds with a 3D printer, pour silicone into them, and make the tires replaceable.

・After several uses, replace them with new tires and test under the same conditions.

■ Mapping and Control Logic

・Mapping: snap all positions to corner coordinates.

・Plates placed in corners make operation difficult due to noise processing.

・Backup functions are important, such as returning to the start position and restarting.

・Memorize the map around the start position and correct position drift when returning.

・Monitor the robot status continuously by wireless communication.

Introduction of the MDD10A Motor Driver Date: 2025.09.15

Changed from TB6612FNG to the MDD10A, which supports a higher current of 10 A.

MDD10A Pin	Teensy Pin	Function
DIR 1	34	Motor 1 direction
PWM 1	33 (PWM)	Motor 1 speed
DIR 2	36	Motor 2 direction
PWM 2	37	Motor 2 speed
GND	GND	Common ground
M1 A/B	Motor 1	Left motor output
M2 A/B	+/-	Right motor output
Power +/-	Battery	Main power input

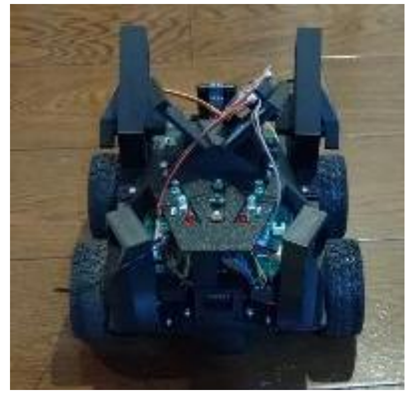
Operation Check Results

· Very stable and accurate.

· It does not wobble when starting or stopping, and stops precisely.

· Four-wheel drive also succeeded by using two motor drivers.

Prototype Robot V3 Development Date: 2025.09.22 to present



Configuration

Microcontroller: Teensy 4.1

Motors: DF Robot Metal RB-Dfr-759 (100 rpm) x4 (four-wheel drive)

Motor drivers: Cytron MDD10A x2 (high-output capable)

Battery: Ni-MH batteries (Eneloop Pro) AA x5 for motors, AA x4 for Teensy

ToF sensors: VL53L0X x6 (front, rear, left, right, and diagonal)

Color sensor: TCS34725

Display: OLED (SSD1306) x1

Frame: created with a 3D printer

DC-DC converter: S7V7F5

AI camera: OpenMV H7 Plus x1 (planned expansion to two cameras in the future)

Buttons: tactile switches x3

LEDs: red, yellow, and green LEDs installed

Voltmeter: for the motor power supply, Teensy power supply, and voltage after the 5 V regulator

Improvements from Prototype Robot V2 to V3

・Motors: JGA25-371 (50 rpm) x2 to DF Robot Metal RB-Dfr-759 (100 rpm) x4 (smaller individual differences).
  Four-wheel drive improved driving performance.
・Motor drivers: TB6612FNG x1 to MDD10A x2 (improved control accuracy).
・Slimmer body: 21 cm to 19.5 cm.
・ToF sensors: added two sensors for diagonal detection, making it possible to detect walls earlier.
・Power supply: changed from Eneloop to Eneloop Pro, increasing the maximum current.
・Voltage regulator: S7V7F5 provides stable 5 V power to the Teensy and AI camera.
・AI camera installed: machine learning improved victim detection capability.
・Buttons installed: pause, restart, and one spare button.
・LEDs installed: three colors, red, yellow, and green, to show victim-detection signals and the robot status.
・Rescue-kit dropping mechanisms installed: two mechanisms, one on each side. The simple mechanism has an almost 100% success rate.
・PCB: integrated into a single board.
・Body: created with a 3D printer.

What Became Possible from Prototype Robot V2 to V3

・The robot can detect victims and drop rescue kits.

Next Tasks (2025.11.30)

・Prototype Robot V3 is still in progress, so complete it by improving the program, installing the gyro sensor, touch sensors, lighting LEDs, and adding another AI camera.
・When everything is powered on, an insufficient power supply was found. The battery and regulator must be changed. Adding a lithium-ion battery is under consideration.
・Build the competition robot by the block tournament.
・Build a spare robot by the block tournament.
・Build a course at the Japan Open scale.